THE RELATIVE FROBENIUS OF AN ETALE MORPHISM IN CHARACTERISTIC P IS AN ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 0CCB: sources/stacks-chow/tags.txt, source lines 15501–15501. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GW: sources/stacks-chow/tags.txt, source lines 3243–3243. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original theorem 025G: sources/stacks-chow/tags.txt, source lines 2813–2813. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $U \to X$ be an étale morphism of schemes where X is a scheme in characteristic p. Then the relative Frobenius $F_{U/X} : U \to U \times_{X,F_X} X$ is an isomorphism.*

Proof. The morphism $F_{U/X}$ is a universal homeomorphism by Varieties, Lemma 0CCB. The morphism $F_{U/X}$ is étale as a morphism between schemes étale over X (Morphisms, Lemma 02GW). Hence $F_{U/X}$ is an isomorphism by Theorem 025G. □

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